Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: O/L Dairy

Physical address of dairy:

22678 W Pioneer RD	Los Banos	Merced	93635
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1930

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X008-X200-X025-XXXX

B. OPERATORS

Lawrence, Henry & Mary

Operator name: Lawrence, Henry & Mary

Telephone no.: (209) 826-0187 (Landline) (Cellular)

22678 W Pioneer RD	Los Banos	CA	93635
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

Lawrence, Henry & Mary

Legal owner name: Lawrence, Henry & Mary

Telephone no.: (209) 826-0187 (Landline) (Cellular)

22678 W Pioneer RD	Los Banos	CA	93635
Mailing Address Number and Street	City	State	Zip Code

This owner is responsible for paying permit fees.

Oliveira, Rose F Trustee

Legal owner name:	Oliveira, Rose F Trustee	Telephone no.:	(209) 000-0000	
			Landline	Cellular
22678 W Pioneer RD		Los Banos	CA	93635
Mailing Address Number and Street		City	State	Zip Code

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	125	75	0
Number under roof	0	0	150	50	50	0
Maximum number	0	0	300	350	350	0
Average number	0	0	150	175	125	0
Avg live weight (lbs)	0	0	800	700		

Predominant milk cow breed: Holstein

Average milk production: 1 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 3,559.83 *tons per reporting period*

Total nitrogen from manure: 37,230.00 *lbs per reporting period*

After ammonia losses (30% loss applied): 26,061.00 *lbs per reporting period*

Total phosphorus from manure: 5,671.19 *lbs per reporting period*

Total potassium from manure: 1.00 *lbs per reporting period*

Total salt from manure: 0.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 5,742,821 *gallons*

Total nitrogen generated: 19,851.84 *lbs*

Total phosphorus generated: 6,029.53 *lbs*

Total potassium generated: 60,766.98 *lbs*

Total salt generated: 180,783.13 *lbs*

	5,742,821	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	5,742,821	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
07/01/2023	Solid commercial fertilizer *UN32*	12.01 *ton*	Dry-weight	0.1	32.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	7,678.71	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	7,678.71	0.00	0.00	0.00

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
D1	8	8	1	none	X083-X017-X026-XXXX
D2	42	42	1	none	X083-X017-X026-XXXX X083-X017-X041-XXXX
D3	19	19	2	manure	X083-X017-X039-XXXX X083-X017-X040-XXXX
Dairy Site #1	54	54	1	process wastewater	X083-X020-X025-XXXX
Dairy Site #2	29	29	2	process wastewater	X083-X020-X025-XXXX
Richard 1	4	4	1	none	X083-X019-X011-XXXX
Richard 2	9	9	1	none	X083-X019-X010-XXXX X083-X019-X011-XXXX
Totals for areas that were used for application	102	102	5		
Totals for areas that were not used for application	63	63	4		
Land application area totals	165	165	9		

B. CROPS AND HARVESTS

D1

Field name: D1

01/01/2023: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 8 Plant date: 01/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
12/31/2023	45.34 *ton*	As-is		10.0	26,200.00	2,300.00	21,900.00		10.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	5.67	296.98	26.07	248.24	1,050.75

D2

Field name: D2

D2

01/01/2023: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 42 Plant date: 01/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
12/31/2023	238.04 *ton*	As-is		10.0	26,200.00	2,300.00	21,900.00		10.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	5.67	296.98	26.07	248.24	1,050.78

D3

Field name: D3

11/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 19 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	320.29 *ton*	As-is		67.2	4,000.00	500.00	4,000.00		9.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.86	134.86	16.86	134.86	1,061.61

06/10/2023: Corn, silage

Crop: Corn, silage Acres planted: 19 Plant date: 06/10/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/06/2023	541.02 *ton*	As-is		67.5	4,200.00	700.00	6,000.00		6.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	28.47	239.19	39.86	341.70	1,110.51

Dairy Site #1

Field name: Dairy Site #1

01/01/2023: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 54 Plant date: 01/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
12/31/2023	306.05 *ton*	As-is		10.0	24,900.00	2,100.00	23,500.00		12.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	5.67	282.25	23.80	266.38	1,244.60

Dairy Site #2

Field name: Dairy Site #2

10/28/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 29 Plant date: 10/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	488.86 *ton*	As-is		58.4	4,500.00	700.00	4,600.00		9.49

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.86	151.72	23.60	155.09	1,330.99

06/09/2023: Corn, silage

Crop: Corn, silage Acres planted: 29 Plant date: 06/09/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/06/2023	825.76 *ton*	As-is		69.2	4,000.00	600.00	7,900.00		9.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	28.47	227.80	34.17	449.90	1,736.49

Richard 1

Field name: Richard 1

10/30/2022: Oats, hay

Crop: Oats, hay Acres planted: 4 Plant date: 10/30/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/27/2023	14.64 *ton*	As-is		10.0	10,900.00	1,300.00	10,900.00		9.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	320.00	52.00	264.00	0.00
Total actual harvest content	3.66	79.79	9.52	79.79	632.45

Richard 2

Field name: Richard 2

10/30/2022: Oats, hay

Crop: Oats, hay Acres planted: 9 Plant date: 10/30/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/27/2023	32.94 *ton*	As-is		10.0	10,900.00	1,300.00	10,900.00		9.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	320.00	52.00	264.00	0.00
Total actual harvest content	3.66	79.79	9.52	79.79	632.45

NUTRIENT BUDGET

A. LAND APPLICATIONS

D1 - 01/01/2023: Alfalfa, hay

Field name: D1

Crop: Alfalfa, hay

Plant date: 01/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	379.39	2,346,480.00 *gal*
Application event totals		0.00	0.00	0.00	379.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	173.89	1,075,470.00 *gal*
Application event totals		0.00	0.00	0.00	173.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	284.54	1,759,860.00 *gal*
Application event totals		0.00	0.00	0.00	284.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	368.85	2,281,300.00 *gal*
Application event totals		0.00	0.00	0.00	368.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	194.96	1,205,830.00 *gal*
Application event totals		0.00	0.00	0.00	194.96	

D2 - 01/01/2023: Alfalfa, hay

Field name: D2

Crop: Alfalfa, hay

Plant date: 01/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	292.81	9,507,710.00 *gal*
Application event totals		0.00	0.00	0.00	292.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	207.76	6,746,130.00 *gal*
Application event totals		0.00	0.00	0.00	207.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	248.65	8,073,750.00 *gal*
Application event totals		0.00	0.00	0.00	248.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	285.04	9,255,560.00 *gal*
Application event totals		0.00	0.00	0.00	285.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	271.12	8,803,400.00 *gal*
Application event totals		0.00	0.00	0.00	271.12	

D3 - 11/01/2022: Oats, silage-soft dough

Field name: D3

Crop: Oats, silage-soft dough

Plant date: 11/01/2022

D3 - 11/01/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/08/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	168.64	42.79	247.30	0.00	70.00 *ton*
Application event totals		168.64	42.79	247.30	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/18/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	327.39	4,809,132.00 *gal*
Application event totals		0.00	0.00	0.00	327.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	252.99	3,716,210.00 *gal*
Application event totals		0.00	0.00	0.00	252.99	

D3 - 06/10/2023: Corn, silage

Field name: D3

Crop: Corn, silage

Plant date: 06/10/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	221.89	3,259,298.00 *gal*
Application event totals		0.00	0.00	0.00	221.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	80.00	0.00	0.00	0.00	
Canal	Surface water	0.00	0.00	0.00	241.02	3,540,300.00 *gal*
Application event totals		80.00	0.00	0.00	241.02	

D3 - 06/10/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	80.00	0.00	0.00	0.00	
Canal	Surface water	0.00	0.00	0.00	352.65	5,180,090.00 *gal*
Application event totals		80.00	0.00	0.00	352.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	252.11	3,703,250.00 *gal*
Application event totals		0.00	0.00	0.00	252.11	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	196.64	2,888,500.00 *gal*
Application event totals		0.00	0.00	0.00	196.64	

Dairy Site #1 - 01/01/2023: Alfalfa, hay

Field name: Dairy Site #1

Crop: Alfalfa, hay

Plant date: 01/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	125.14	40.62	408.29	1,139.12	1,321,001.00 *gal*
Application event totals		125.14	40.62	408.29	1,139.12	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	90.90	29.51	296.58	827.46	959,580.00 *gal*
Application event totals		90.90	29.51	296.58	827.46	

Dairy Site #1 - 01/01/2023: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	245.53	10,250,400.00 *gal*
Application event totals		0.00	0.00	0.00	245.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	209.29	8,737,530.00 *gal*
Application event totals		0.00	0.00	0.00	209.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	223.10	9,313,860.00 *gal*
Application event totals		0.00	0.00	0.00	223.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	259.54	10,835,300.00 *gal*
Application event totals		0.00	0.00	0.00	259.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	177.12	7,394,490.00 *gal*
Application event totals		0.00	0.00	0.00	177.12	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	237.97	9,934,790.00 *gal*
Application event totals		0.00	0.00	0.00	237.97	

Dairy Site #2 - 10/28/2022: Oats, silage-soft dough

Dairy Site #2 - 10/28/2022: Oats, silage-soft dough

Field name: Dairy Site #2

Crop: Oats, silage-soft dough

Plant date: 10/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/12/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	12.73	3.31	27.21	169.44	892,160.00 *gal*
Canal	Surface water	0.00	0.00	0.00	198.81	4,457,408.00 *gal*
Application event totals		12.73	3.31	27.21	368.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	99.34	32.25	324.11	904.25	563,155.00 *gal*
Canal	Surface water	0.00	0.00	0.00	167.06	3,745,582.00 *gal*
Application event totals		99.34	32.25	324.11	1,071.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	74.40	24.15	242.74	677.24	421,775.00 *gal*
Canal	Surface water	0.00	0.00	0.00	140.64	3,153,228.00 *gal*
Application event totals		74.40	24.15	242.74	817.88	

Dairy Site #2 - 06/09/2023: Corn, silage

Field name: Dairy Site #2

Crop: Corn, silage

Plant date: 06/09/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	53.06	9.75	104.60	454.77	878,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	312.24	7,000,422.00 *gal*
Application event totals		53.06	9.75	104.60	767.01	

Dairy Site #2 - 06/09/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	80.00	0.00	0.00	0.00	
Canal	Surface water	0.00	0.00	0.00	243.29	5,454,540.00 *gal*
Application event totals		80.00	0.00	0.00	243.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	80.00	0.00	0.00	0.00	
Canal	Surface water	0.00	0.00	0.00	200.04	4,484,987.00 *gal*
Application event totals		80.00	0.00	0.00	200.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	42.73	7.85	84.24	366.28	707,150.00 *gal*
Canal	Surface water	0.00	0.00	0.00	238.20	5,340,475.00 *gal*
Application event totals		42.73	7.85	84.24	604.48	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	190.73	4,276,295.00 *gal*
Application event totals		0.00	0.00	0.00	190.73	

Richard 1 - 10/30/2022: Oats, hay

Field name: Richard 1

Crop: Oats, hay

Plant date: 10/30/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Richard 1 - 10/30/2022: Oats, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/10/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	209.96	649,276.00 *gal*
Application event totals		0.00	0.00	0.00	209.96	

03/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	170.17	526,236.00 *gal*
Application event totals		0.00	0.00	0.00	170.17	

Richard 2 - 10/30/2022: Oats, hay

Field name: Richard 2

Crop: Oats, hay

Plant date: 10/30/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/10/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	209.96	1,460,871.00 *gal*
Application event totals		0.00	0.00	0.00	209.96	

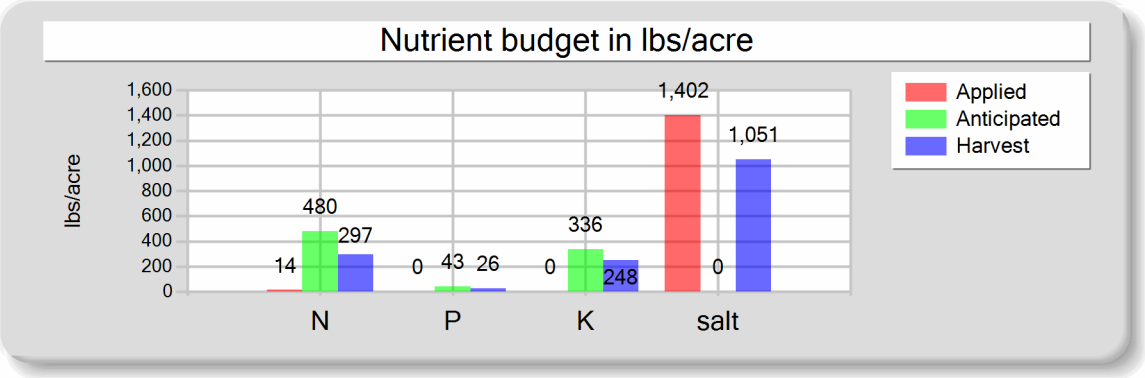
03/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	170.17	1,184,031.00 *gal*
Application event totals		0.00	0.00	0.00	170.17	

B. NUTRIENT BUDGET

D1 - 01/01/2023: Alfalfa, hay

Field name: D1 Crop: Alfalfa, hay Plant date: 01/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,401.63
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	14.00	0.00	0.00	1,401.63
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	296.98	26.07	248.24	1,050.75
Nutrient balance	-282.98	-26.07	-248.24	350.88
Applied to removed ratio	0.05	0.00	0.00	1.33

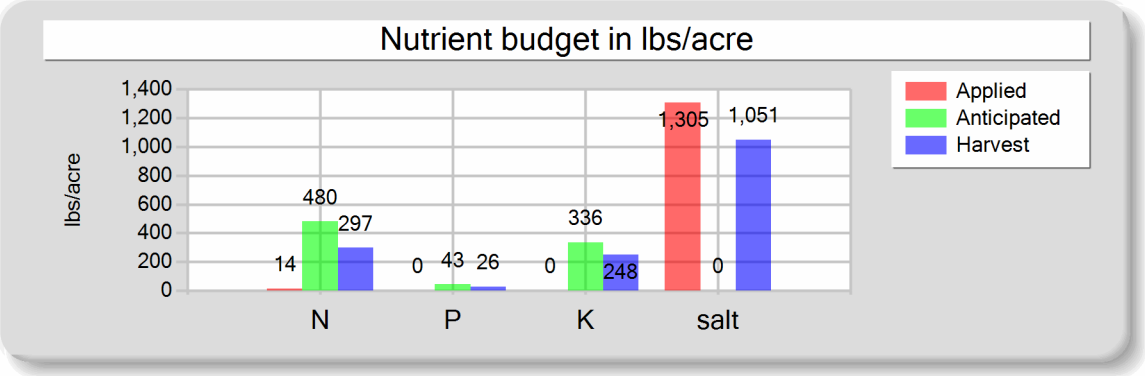
Fresh water applied	
8,668,940.00	*gallons*
319.25	*acre-inches*
39.91	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

D2 - 01/01/2023: Alfalfa, hay

Field name: D2 Crop: Alfalfa, hay Plant date: 01/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,305.38
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	14.00	0.00	0.00	1,305.38
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	296.98	26.07	248.24	1,050.78
Nutrient balance	-282.98	-26.07	-248.24	254.60
Applied to removed ratio	0.05	0.00	0.00	1.24

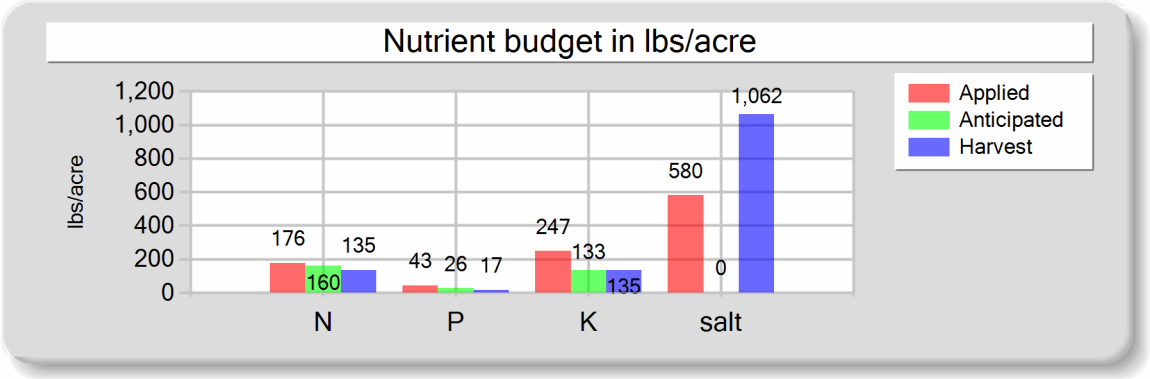
Fresh water applied	
42,386,550.00	*gallons*
1,560.95	*acre-inches*
37.17	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

D3 - 11/01/2022: Oats, silage-soft dough

Field name: D3 Crop: Oats, silage-soft dough Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	168.64	42.79	247.30	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	580.39
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	175.64	42.79	247.30	580.39
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	134.86	16.86	134.86	1,061.61
Nutrient balance	40.78	25.93	112.44	-481.22
Applied to removed ratio	1.30	2.54	1.83	0.55

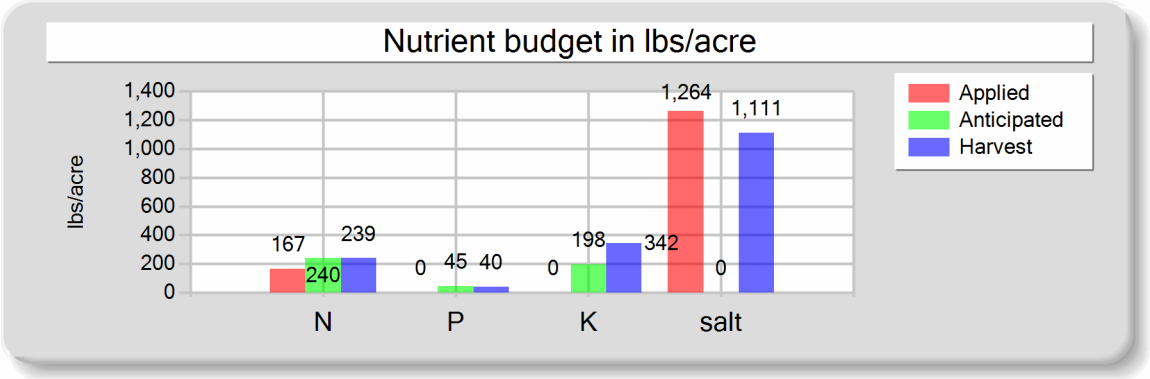
Fresh water applied	
8,525,342.00	*gallons*
313.96	*acre-inches*
16.52	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

D3 - 06/10/2023: Corn, silage

Field name: D3 Crop: Corn, silage Plant date: 06/10/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	160.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,264.30
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	167.00	0.00	0.00	1,264.30
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	239.19	39.86	341.70	1,110.51
Nutrient balance	-72.19	-39.86	-341.70	153.78
Applied to removed ratio	0.70	0.00	0.00	1.14

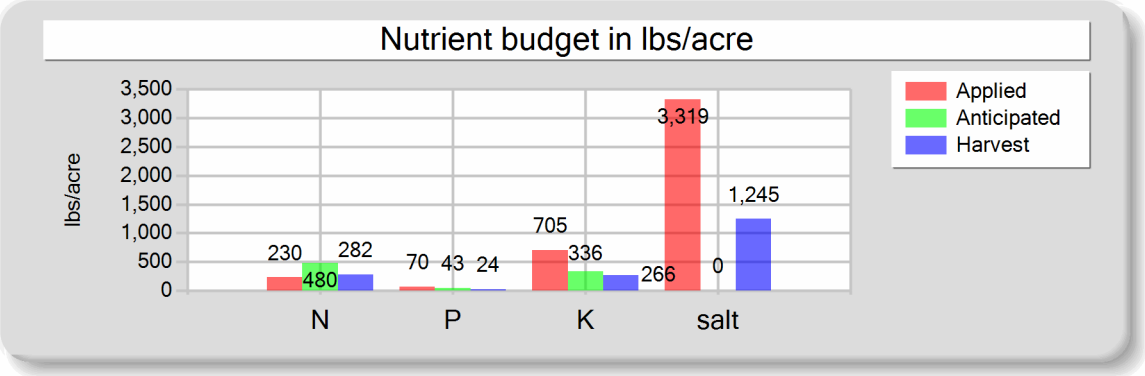
Fresh water applied	
18,571,438.00	*gallons*
683.92	*acre-inches*
36.00	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Dairy Site #1 - 01/01/2023: Alfalfa, hay

Field name: Dairy Site #1 Crop: Alfalfa, hay Plant date: 01/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	216.04	70.13	704.87	1,966.58
Fresh water	0.00	0.00	0.00	1,352.55
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	230.04	70.13	704.87	3,319.14
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	282.25	23.80	266.38	1,244.60
Nutrient balance	-52.20	46.33	438.49	2,074.53
Applied to removed ratio	0.82	2.95	2.65	2.67

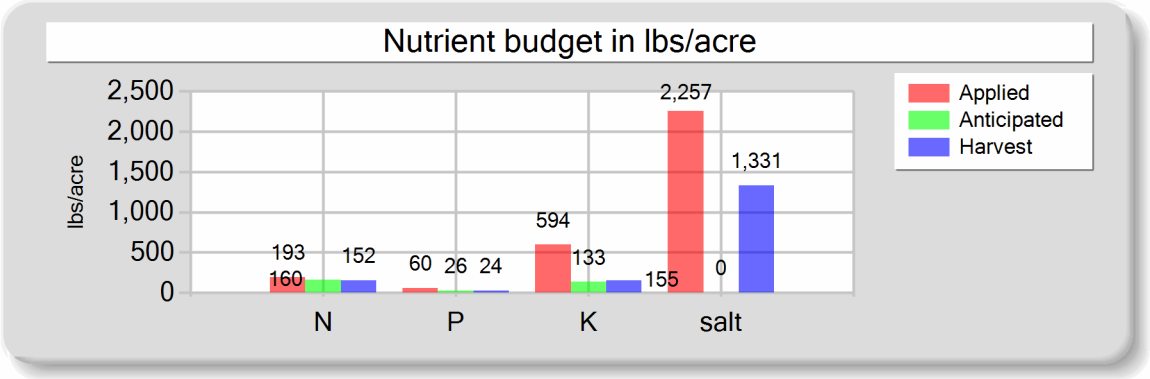
Fresh water applied	
56,466,370.00	*gallons*
2,079.46	*acre-inches*
38.51	*inches/acre*

Process wastewater applied	
2,280,581.00	*gallons*
83.99	*acre-inches*
1.56	*inches/acre*

Total harvests for the crop	
1	*harvests*

Dairy Site #2 - 10/28/2022: Oats, silage-soft dough

Field name: Dairy Site #2 Crop: Oats, silage-soft dough Plant date: 10/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	186.47	59.71	594.06	1,750.93
Fresh water	0.00	0.00	0.00	506.52
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	193.47	59.71	594.06	2,257.45
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	151.72	23.60	155.09	1,330.99
Nutrient balance	41.76	36.11	438.97	926.46
Applied to removed ratio	1.28	2.53	3.83	1.70

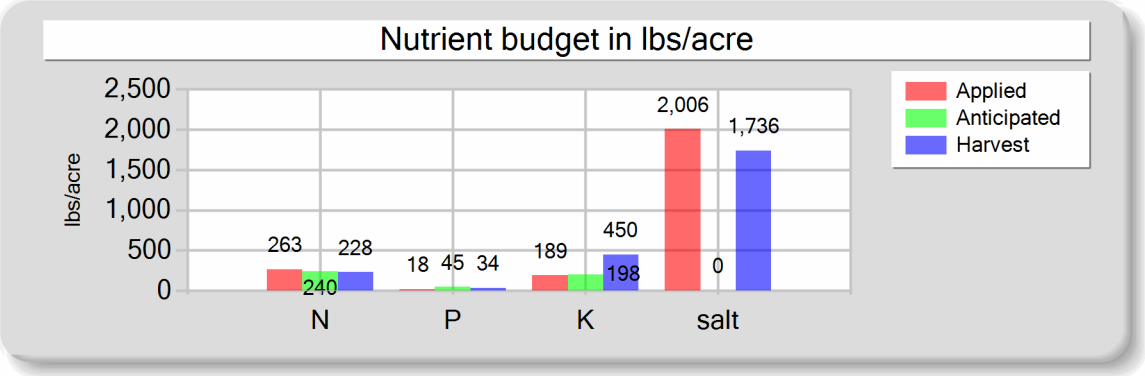
Fresh water applied	
11,356,218.00	*gallons*
418.21	*acre-inches*
14.42	*inches/acre*

Process wastewater applied	
1,877,090.00	*gallons*
69.13	*acre-inches*
2.38	*inches/acre*

Total harvests for the crop	
1	*harvests*

Dairy Site #2 - 06/09/2023: Corn, silage

Field name: Dairy Site #2 Crop: Corn, silage Plant date: 06/09/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	160.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	95.79	17.61	188.84	821.05
Fresh water	0.00	0.00	0.00	1,184.50
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	262.79	17.61	188.84	2,005.55
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	227.80	34.17	449.90	1,736.49
Nutrient balance	34.99	-16.56	-261.05	269.06
Applied to removed ratio	1.15	0.52	0.42	1.15

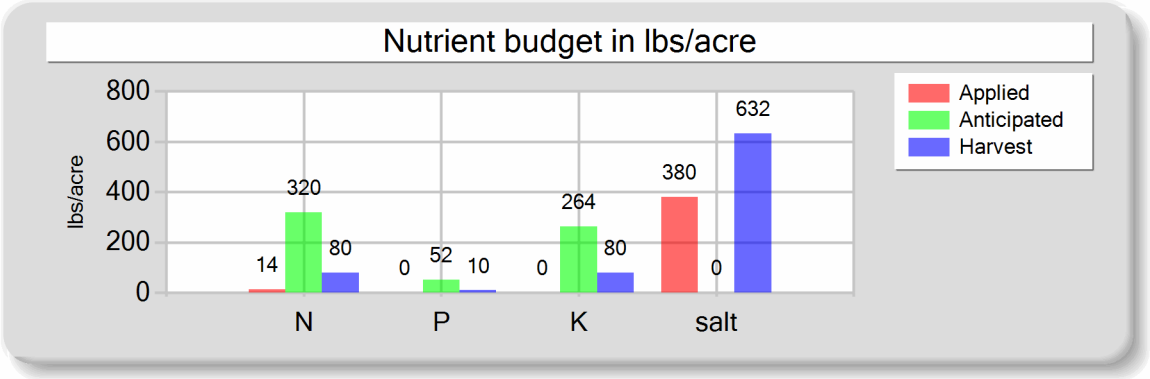
Fresh water applied	
26,556,719.00	*gallons*
977.99	*acre-inches*
33.72	*inches/acre*

Process wastewater applied	
1,585,150.00	*gallons*
58.38	*acre-inches*
2.01	*inches/acre*

Total harvests for the crop	
1	*harvests*

Richard 1 - 10/30/2022: Oats, hay

Field name: Richard 1 Crop: Oats, hay Plant date: 10/30/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	380.12
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	14.00	0.00	0.00	380.12
Anticipated crop nutrient removal	320.00	52.00	264.00	0.00
Actual crop nutrient removal	79.79	9.52	79.79	632.45
Nutrient balance	-65.79	-9.52	-79.79	-252.32
Applied to removed ratio	0.18	0.00	0.00	0.60

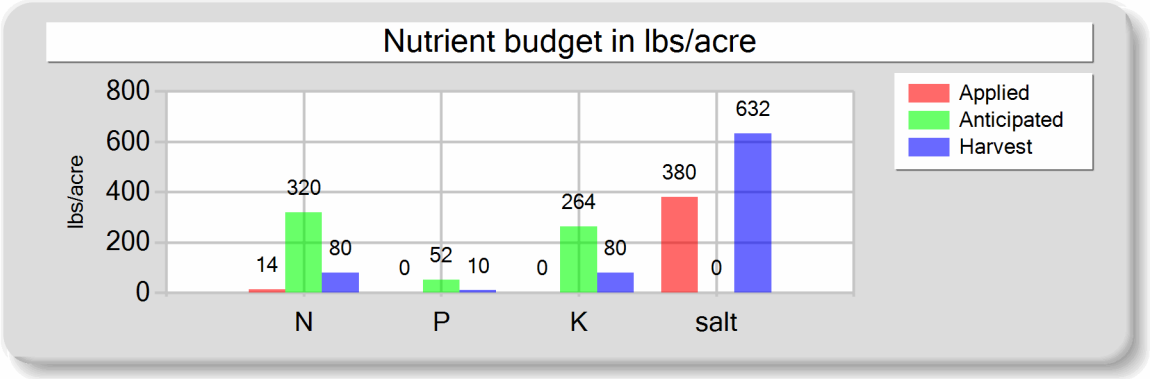
Fresh water applied	
1,175,512.00	*gallons*
43.29	*acre-inches*
10.82	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Richard 2 - 10/30/2022: Oats, hay

Field name: Richard 2 Crop: Oats, hay Plant date: 10/30/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	380.12
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	14.00	0.00	0.00	380.12
Anticipated crop nutrient removal	320.00	52.00	264.00	0.00
Actual crop nutrient removal	79.79	9.52	79.79	632.45
Nutrient balance	-65.79	-9.52	-79.79	-252.32
Applied to removed ratio	0.18	0.00	0.00	0.60

Fresh water applied	
2,644,902.00	*gallons*
97.40	*acre-inches*
10.82	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Fall 2022 Solid Manure

Sample and source description: Fall 2022 Solid Manure

Sample date: 10/14/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 14.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,800.00	6,800.00	39,300.00							
DL	100.00	100.00	100.00							

Spring Solid Manure

Sample and source description: Spring Solid Manure

Sample date: 06/01/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 18.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,000.00	4,800.00	26,400.00							
DL	100.00	100.00	100.00							

Fall Solid Manure

Sample and source description: Fall Solid Manure

Sample date: 10/06/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 13.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,500.00	2,200.00	15,700.00							
DL	100.00	100.00	100.00							

B. PROCESS WASTEWATER ANALYSES

2022 WWSQ4

Sample and source description: 2022 WWSQ4

Sample date: 11/18/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	49.60	27.20	0.00	0.00	12.90	106.00								1,100.00	660
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ1

Sample and source description: WWSQ1

Sample date: 03/24/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	613.00	342.00	0.00	0.00	199.00	2,000.00								5,800.00	5,580
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ2

Sample and source description: WWSQ2

Sample date: 06/01/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	33.60	14.00	0.00	0.00	7.51	66.30								751.00	260
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ3

Sample and source description: WWSQ3

Sample date: 10/06/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	210.00	157.00	0.00	0.00	38.60	414.00								3,520.00	1,800
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ4

Sample and source description: WWSQ4

Sample date: 12/21/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	64.40	16.80	0.00	0.00	17.40	225.00								1,440.00	1,060
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

Canal

Canal

Sample description: Canal

Sample date: 07/25/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								228.00	155
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

D1

Soil

Sample and source description: Soil

Sample date: 12/19/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			22.00				
DL			4.00				

D2

D2

Soil

Sample and source description: Soil

Sample date: 12/19/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			22.00				
DL			4.00				

D3

Soil

Sample and source description: Soil

Sample date: 12/19/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			22.00				
DL			4.00				

Dairy Site #1

Soil

Sample and source description: Soil

Sample date: 12/19/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			23.00				
DL			4.00				

Dairy Site #2

Dairy Site #2

Soil

Sample and source description: Soil

Sample date: 12/19/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			25.00				
DL			4.00				

E. PLANT TISSUE ANALYSES

D1 - 01/01/2023: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 07/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 10.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,200.00	2,300.00	21,900.00		10.30
DL	100.00	100.00	10.00		0.01

D2 - 01/01/2023: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 07/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 10.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,200.00	2,300.00	21,900.00		10.30
DL	100.00	100.00	100.00		0.01

D3 - 11/01/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,000.00	500.00	4,000.00		9.60
DL	100.00	100.00	100.00		0.01

D3 - 06/10/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,200.00	700.00	6,000.00		6.00
DL	100.00	100.00	100.00		0.01

Dairy Site #1 - 01/01/2023: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 07/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 10.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,900.00	2,100.00	23,500.00		12.20
DL	100.00	100.00	100.00		0.01

Dairy Site #2 - 10/28/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 58.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,500.00	700.00	4,600.00		9.49
DL	100.00	100.00	100.00		0.01

Dairy Site #2 - 06/09/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,000.00	600.00	7,900.00		9.90
DL	100.00	100.00	100.00		0.01

Richard 1 - 10/30/2022: Oats, hay

Oat Hay

Sample and source description: Oat Hay

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 10.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,900.00	1,300.00	10,900.00		9.60
DL	100.00	100.00	100.00		0.01

Richard 2 - 10/30/2022: Oats, hay

Oat Hay

Sample and source description: Oat Hay

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 10.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,900.00	1,300.00	10,900.00		9.60
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

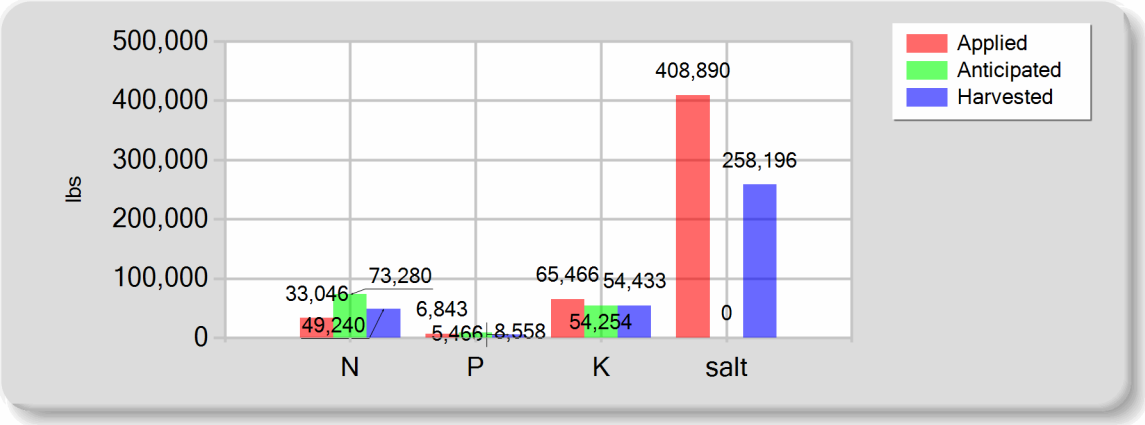
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

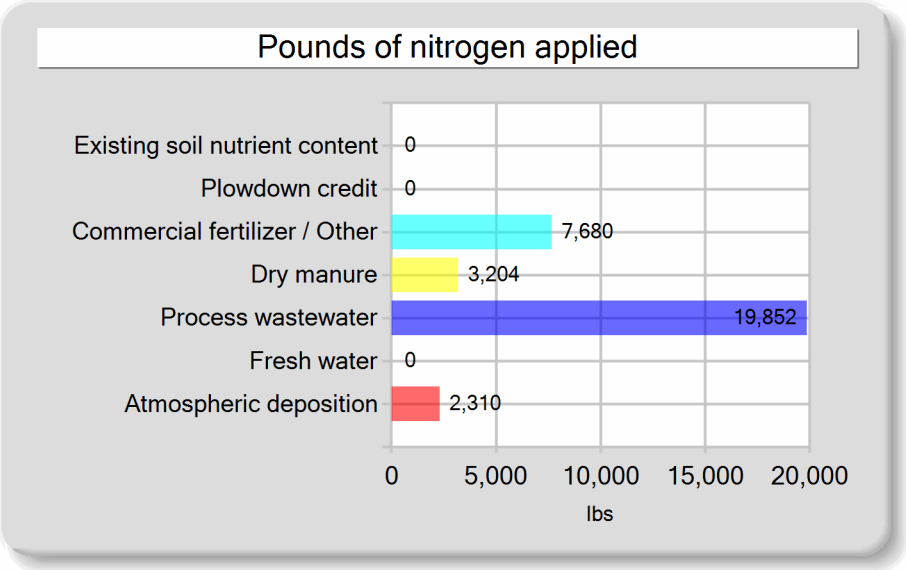
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

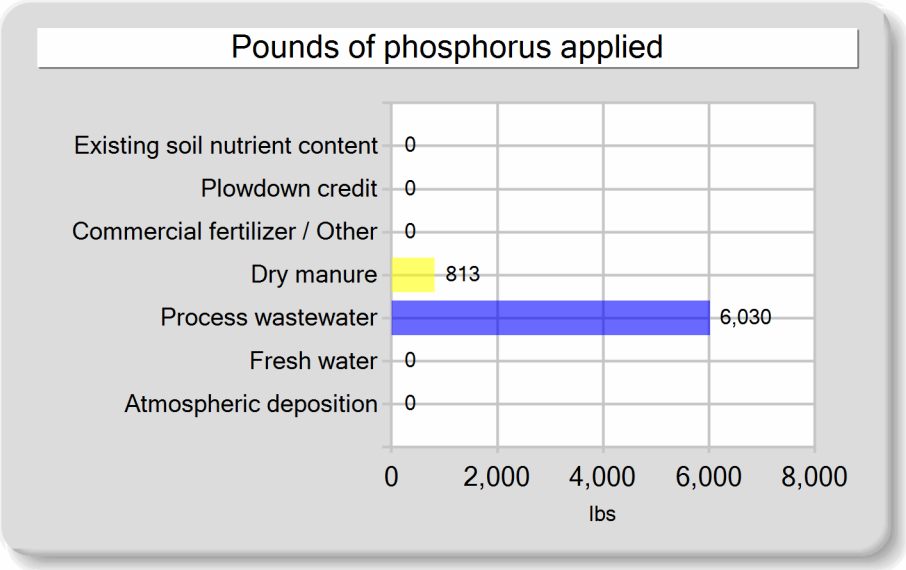
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	7,680.00	0.00	0.00	0.00
Dry manure	3,204.21	813.01	4,698.71	0.00
Process wastewater	19,851.84	6,029.53	60,766.98	180,783.13
Fresh water	0.00	0.00	0.00	228,106.89
Atmospheric deposition	2,310.00	0.00	0.00	0.00
Total nutrients applied	33,046.05	6,842.54	65,465.69	408,890.03
Anticipated crop nutrient removal	73,280.00	8,557.60	54,254.40	0.00
Actual crop nutrient removal	49,240.35	5,465.70	54,432.72	258,196.39
Nutrient balance	-16,194.30	1,376.84	11,032.97	150,693.64
Applied to removed ratio	0.67	1.25	1.20	1.58

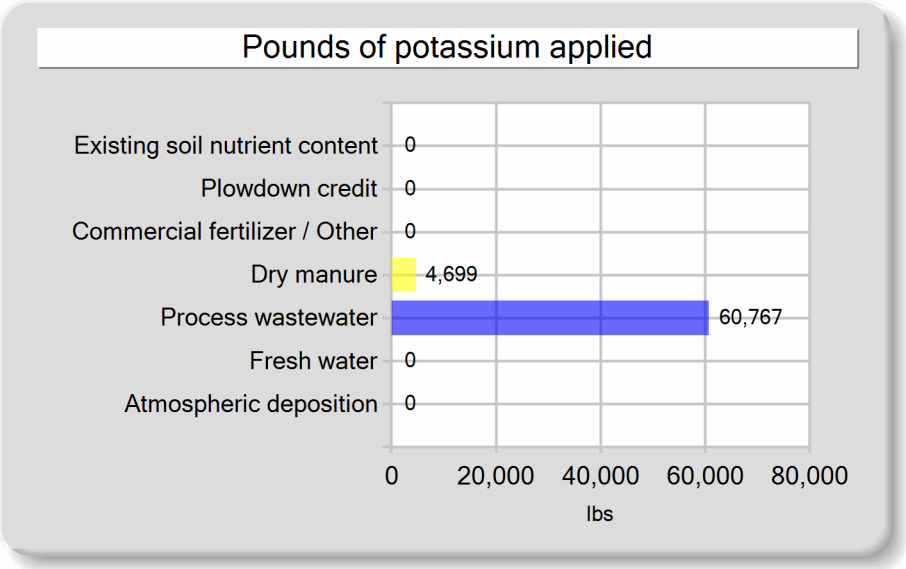
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

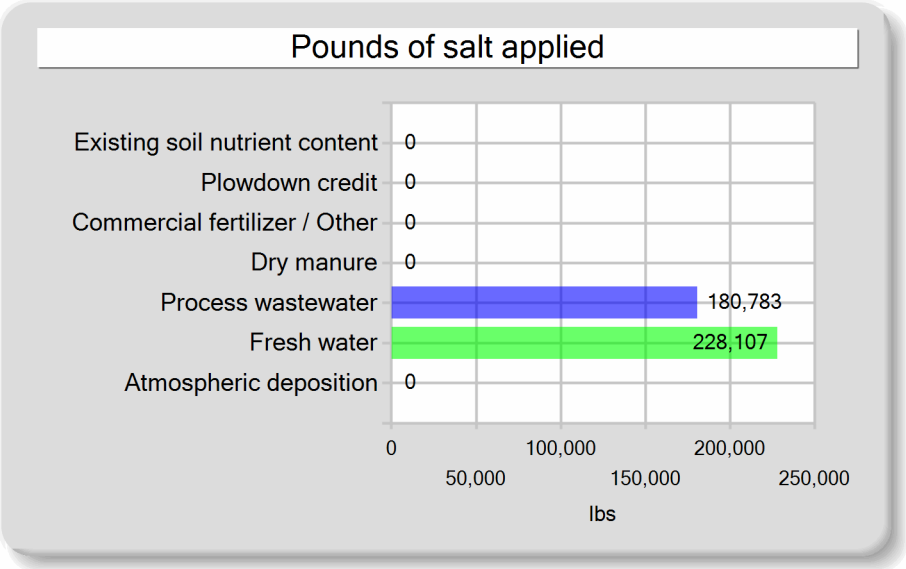


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Facility was depopulated in July of 2023. Facility remains empty as of June 1st, 2024.

Richards 1 and 2 remained fallow during corn silage season.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Mary Lawrence (Jun 13, 2024 12:40 PDT)	Mary Lawrence (Jun 13, 2024 12:40 PDT)
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Henry & Mary Lawrence	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jun 13, 2024	Jun 13, 2024
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

DENELE ANALYTICAL, INC.
1232 SOUTH AVENUE
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: O/L DAIRY
Mailing address: 22678 Pioneer Rd
(not facility address) Los Banos
Phone#: 209-704-0187

Authorized Copy Released To: F+R
Sampled By:
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
H1 EC*, NO3-N, NH3-N*
H6 EC, TN, TDS
H7 Combination of H1 & H6
H14 EC, NO3-N, NH3-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
H3 EC*, NH4-N, TKN, TP, K, TDS
H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
H8 Combination of H3 &H5
H9 Combination of H4 &H5

TILE DRAIN
H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*
___ (Check here if performed)

DISCHARGE WATER
E1 EC*, Temp*, pH*, NO3-N, NH3-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
E2 EC,* Temp*, pH*, NO3-N, NH3-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
E3 EC*, Temp*, pH*, NO3-N, NH3-N*, TP, Total & Fecal Coliforms
E4 EC*, Temp*, pH*, NO3-N, NH3-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
M1 % Moisture
M3 TN TP, K, % Moisture
M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
M7 Combination of M3 & M4

PLANT TISSUE
F18 % Moisture, Ash, TN, P, K, DM and as received
F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
K3 Soluble Phosphorus
K4 1ft NO3, Organic Matter
K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
K7 Combination of K3 & K4
K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	7/19/23	7:00 Am	CANAL WATER	~~22.4~~ AM	15.0		X				
2	7/19/23	9:00 Am	Domestic Well	~~24.0~~ AM	24.4		X				
3	7/19/23	9:00 Am	Dairy-1 Dairy-1 Alfalfa							X	
4	7/19/23	9:15 Am	D-2 D-2 Alfalfa							X	
5											
6											
7											
8											
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	*(LABORATORY USE ONLY)* Logged In By:	LABORATORY NUMBER
1st	[signature]					
2nd	[signature]	DENELE ANALYTICAL, INC.	7-19-23 @ 11:36			T3 200009

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
O & L DAIRY
22678 W. Pioneer Rd.
Los Banos, CA 93635

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 7/19/23 11:36 AM
SUBMITTED BY: Mary
ANALYZED DATE: 7/19/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 24.4c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Domestic Well	T3200009B	Electrical Conductivity (EC)	EPA 120.1 and	1.35	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	6.91	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	1.09	mg/L
		Soluble Salts (SALT-SOL)	Calculation	863	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	90.0	91.0 (T3199030A)	91.0 (T3199030A)	0.000
Electrical Conductivity (EC)	97.9			0.635
Nitrate Nitrogen (NO3-N)	101	98.5 (T3199027A)	97.0 (T3199027A)	1.45

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

2023 Annual Report-O & L

Final Audit Report 2024-06-13

Created:	2024-06-13
By:	Kristen Jones (kjones@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAAZWtQiR8nqdcgM3FFOnazmv1rBZzqlxIr

"2023 Annual Report-O & L" History

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Signer cowsr4us@gmail.com entered name at signing as Mary Lawrence
2024-06-13 - 7:40:12 PM GMT

Document e-signed by Mary Lawrence (cowsr4us@gmail.com)
Signature Date: 2024-06-13 - 7:40:14 PM GMT - Time Source: server

Agreement completed.
2024-06-13 - 7:40:14 PM GMT

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